

Om Shah

Full Stack Software Engineer

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SUMMARY:

Full Stack Software Engineer with 3+ years of experience building AI-enabled cloud-native platforms using Python, FastAPI, React, Kubernetes, and distributed microservices architectures. Strong expertise in LLM applications, RAG pipelines, Agentic AI workflows, Kafka-based event streaming, and scalable backend engineering across AWS environments. Proven track record of optimizing system throughput, reducing latency, and delivering production-grade AI-powered enterprise applications.

SKILLS:

Programming Languages: Python, Java, JavaScript, TypeScript, SQL
Backend & Microservices: FastAPI, Django, Flask, REST APIs, GraphQL, gRPC, Microservices Architecture, Event-Driven Systems, API Gateway, JWT, OAuth2
Frontend Development: React, Next.js, Redux, Tailwind CSS, HTML5, CSS3
LLM, RAG & Agentic AI: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, Vector Databases, Semantic Search, Prompt Engineering, Agentic AI Workflows, OpenAI APIs, Hugging Face Transformers, AI Orchestration Pipelines
AI & Data Engineering: NLP, Machine Learning, Pandas, NumPy, ETL Pipelines, Real-Time Data Processing, Apache Kafka
Databases & Storage: PostgreSQL, MongoDB, MySQL, Redis, DynamoDB
Cloud & DevOps: AWS (EC2, S3, Lambda, RDS, API Gateway), Docker, Kubernetes, Terraform, GitHub Actions, CI/CD Pipelines
Observability & Testing: Grafana, ELK Stack, CloudWatch, PyTest, Selenium, Jest, Integration Testing
Architecture & Systems: Distributed Systems, Scalability Engineering, Load Balancing, Caching Strategies

EDUCATION:

Master's in Information Technology Arizona State University, USA	May 2026
Bachelor of Engineering in Information Technology Gujarat Technological University, India	May 2024

EXPERIENCE:

ServiceNow, USA | Nov 2025 – Current | Full Stack Software Engineer

- Architected AI-enabled microservices using FastAPI, React, PostgreSQL, and Kubernetes, increasing enterprise platform throughput by 45% while reducing API response latency by 38% across distributed cloud environments.
- Engineered Retrieval-Augmented Generation (RAG) pipelines leveraging vector embeddings, semantic search, and OpenAI APIs, improving enterprise knowledge retrieval relevance by 42% and accelerating search response performance.
- Developed Agentic AI automation workflows for intelligent ticket triage, contextual reasoning, and workflow orchestration, reducing manual operational effort by 45% across internal support systems.
- Built scalable Kafka-driven event streaming architectures integrated with Redis caching and asynchronous processing, improving real-time data processing efficiency by 50% while lowering backend service load.
- Designed inference-ready LLM integration services using FastAPI, Docker, and Kubernetes to support AI-powered recommendation engines and enterprise automation systems.
- Implemented AI-assisted observability pipelines utilizing Grafana, ELK Stack, and anomaly detection logic to identify failures across the distributed microservices infrastructure proactively.

Infinite Infolab, India | Feb 2022 – Jul 2024 | Software Engineer

- Developed scalable backend platforms using Django, Flask, FastAPI, and PostgreSQL, enabling high-volume API integrations across 10+ enterprise applications and improving system interoperability by 40%.
- Engineered distributed microservices architectures with RabbitMQ and Celery-based asynchronous processing, supporting high-throughput background task execution and AI inference workloads.
- Built AI-assisted data processing pipelines using Python, Pandas, and NLP preprocessing techniques, improving analytical accuracy and reducing manual validation effort across enterprise reporting systems.
- Designed responsive React and Redux-based frontend applications integrated with real-time APIs, increasing user retention by 28% and reducing bounce rates by 25%.
- Implemented cloud-native deployments using Docker, Kubernetes, AWS EC2, and CI/CD pipelines, improving deployment reliability and accelerating release cycles for production systems.
- Developed intelligent automation services integrating NLP-driven workflows, semantic processing, and real-time streaming pipelines to streamline enterprise operational tasks.

PROJECTS:

AI-Powered Customer Support Platform

- Built a production-style AI-powered customer support platform integrating LLM-based conversational workflows, real-time messaging, and Kafka-driven event streaming across distributed services.
- Designed scalable microservices with Redis caching and PostgreSQL-backed workflows capable of supporting concurrent enterprise support operations with low-latency communication.
- Implemented JWT-secured authentication and containerized Kubernetes deployments, improving scalability, fault tolerance, and deployment consistency for cloud-native environments.

LLM-Powered Semantic Q&A Search Engine

- Developed an LLM-powered semantic search engine leveraging vector embeddings and Retrieval-Augmented Generation (RAG) architecture for contextual enterprise question answering.
- Optimized inference and retrieval workflows to support 5K+ concurrent search requests under sub-500ms response latency across distributed backend services.
- Engineered scalable FastAPI-based APIs for intelligent document retrieval, contextual ranking, and AI-enhanced knowledge discovery.

Cloud-Native Full Stack E-Commerce Platform

- Built a scalable full-stack e-commerce platform using React, FastAPI, PostgreSQL, and AWS, supporting secure user authentication, product catalog management, and real-time order processing workflows.
- Engineered microservices-based backend APIs with Redis caching and asynchronous processing, improving transaction handling performance and reducing API response latency across high-traffic operations.
- Implemented Dockerized Kubernetes deployments with CI/CD automation using GitHub Actions, enabling zero-downtime releases, infrastructure scalability, and reliable production monitoring.

CERTIFICATIONS:

Machine Learning for Natural Language Processing
Python Programming